# RP-GHOST

## An Oracle-Seeded Cybernetic Computational Architecture & Theoretical Evaluation

**Author:** George Wagenknecht

**Domain:** Cybernetics · Optoelectronics · Information Theory

**Type:** Full Technical Evaluation & System Specification

**ABSTRACT**

RP-GHOST is an experimental cybernetic architecture combining camera-based environmental sensing, optoelectronic logic, photon-electron statistical modeling, temporal coordination, and independent computational evaluation. The physical device uses an optical acquisition system (USB camera or IR laser-reflector bench) to observe environmental and quantum fluctuations. Changes are processed through parallel probabilistic (Driver A) and deterministic (Driver B) decision pathways. Their outputs are combined via charge-integrating threshold links, logic gates, and 74HC74 clocked state retention mechanisms. This paper presents a complete evaluation spanning computational proof-of-work search complexity, optoelectronic photon statistics, single-photon emission models, and physical feedback protocols.

## 1. COMPUTATIONAL COMPLEXITY & PROOF-OF-WORK ANALYSIS

The core distinction in evaluating the architecture's mining and search performance is **exponential search complexity vs. oracle processing time**. The oracle can generate its 64-bit input in linear time `O(n)`, but Bitcoin SHA-256d proof-of-work still requires finding a nonce whose hash falls below the network target. The codebase explicitly separates these two operations.

### 1.1 Difficulty and Search Time Scaling

The configuration specifies local testing difficulty via:

```
TEST_DIFFICULTY = 2 ** (TEST_BITS - 32)
```

This converts the requested number of leading-zero-equivalent bits into Bitcoin difficulty. The expected number of hashes required to satisfy a target defined by *k* bits is given by:

$$E[H] = 2^{TEST_BITS}$$

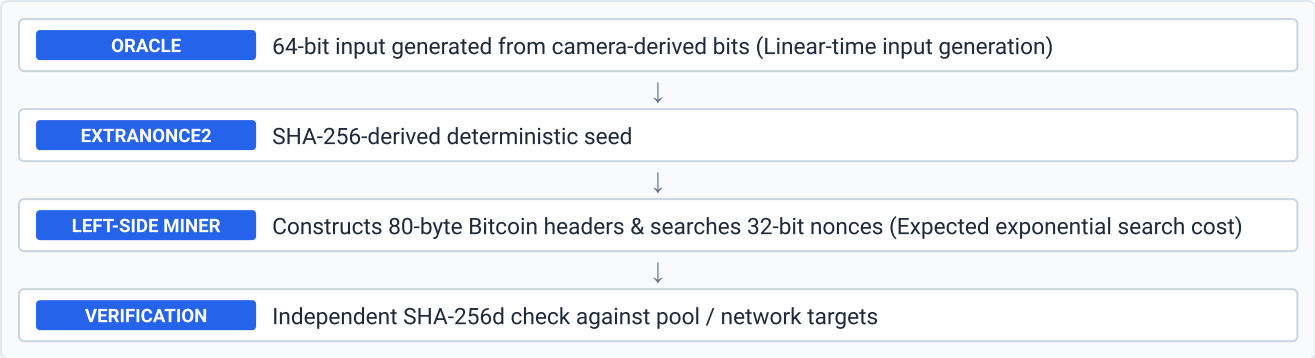
Assuming a processing baseline of 500,000 SHA-256d hashes per second, search times scale exponentially as follows:

| Test Bits | Expected Hashes (E[H]) | Expected Search Duration |
|---|---|---|
| 20 bits | 1,048,576 | 2.1 seconds |
| 24 bits | 16,777,216 | 33.6 seconds |
| 28 bits | 268,435,456 | 8.9 minutes |
| 32 bits | 4,294,967,296 | 2.4 hours |
| 40 bits | 1,099,511,627,776 | 25.5 days |
| 47 bits | 140,737,488,355,328 | 8.9 years |
| 48 bits | 281,474,976,710,656 | 17.8 years |

*Note: These represent mathematical expectation values, not guaranteed completion upper bounds.*

## 2. WHY THE ORACLE CANNOT SOLVE THE COMPLEXITY DIFFERENCE

The system architecture separates seed generation from header search. The oracle determines the starting input, not the probability of satisfying the mathematical target:




Even if the oracle generates its input instantaneously (in `O(n)` linear time), a random SHA-256d hash output retains an identical probability of satisfying a given difficulty target.

## 3. LOCAL TEST MODE VS. PRODUCTION TARGET VERIFICATION

The implementation incorporates a local testing override mechanism:

```
TEST_MODE = True
TEST_BITS = 48

# Target selection logic
return max(
    difficulty_to_target(TEST_DIFFICULTY),
    cls.pool_share_target(job)
)
```

This logic compares local testing thresholds against actual pool thresholds and selects the easier target. Consequently:

- Local test shares can be found relatively quickly when difficulty parameters are reduced.
- Actual pool shares must strictly satisfy the pool's real target.
- Network blocks must satisfy the Bitcoin network target.
- Local test shares are not valid Bitcoin pool shares merely because the local verifier accepts them.

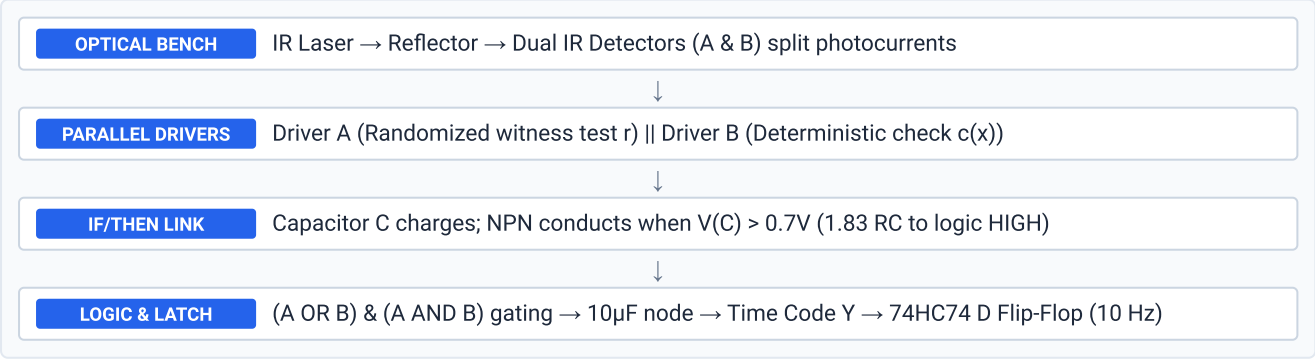
## 4. ARCHITECTURAL LIMITATION & MATHEMATICAL FORMULATION

The oracle functions as a physical seed generator rather than a proof-of-work accelerator. Linear-time oracle operation does not imply linear-time SHA-256d search complexity:

$$T_{total} = T_{oracle} + T_{search} \quad \text{where} \quad T_{oracle} = O(n) \quad \text{and} \quad E[T_{search}] \propto 2^{k}$$

## 5. OPTOELECTRONIC RP DECISION CIRCUIT ARCHITECTURE

The physical device models a hardware realization of a Randomized Polynomial-time (RP) decision procedure with one-sided error. An IR laser emits light across an optical bench to two detectors (A and B), driving parallel execution paths.




### 5.1 Threshold Dynamics & Transistor Kinetics

Driver B charges capacitor *C* through drive resistance *R*. The voltage follows:

$$V(C) = V_s (1 - e^{-t / RC})$$

For $V_s$ = 5V, conduction begins at V(C) = 0.7V (t = 0.15 RC). However, standard 74HC logic input requires $V_{in} \approx 3.5V$ for logic HIGH, reaching valid input voltage at **1.83 RC**. The capacitor integrates charge across N = (C · 0.7V)/e electrons, reducing relative fluctuations to 1 / √N (≈ 0.05% for C = 1 pF), establishing deterministic switching.

## 6. PHOTON STATISTICS & ELECTRON REGULATING MECHANICS

Manipulating electron streams directly alters photon emission statistics. The correlation between electron Fano factor $F_e$ and detected photon Fano factor $F_{ph}$ under quantum efficiency η is governed by:

$$F_{ph} = 1 - \eta(1 - F_e)$$

| Regime | Electron Statistics ($F_e$) | Photon Statistics ($F_{ph}$) | Mandel Q / $g^{(2)}(0)$ Behavior |
|---|---|---|---|
| **Random (Driver A)** | $F_e = 1$ (Poissonian) | $F_{ph} = 1$ (Poissonian) | Q = 0, $g^{(2)}(0) = 1$ (Standard laser shot noise) |
| **Regulated (Driver B)** | $F_e = 0$ (Sub-Poissonian) | $F_{ph} = 1 - \eta$ (Sub-Poissonian) | Q < 0, $g^{(2)}(0) < 1$ (Antibunched single-photon emission) |
| **Bunched (Avlanche/Traps)** | $F_e > 1$ (Super-Poissonian) | $F_{ph} > 1$ (Super-Poissonian) | Q > 0, $g^{(2)}(0) > 1$ (Thermal / bunched photon arrivals) |

## 7. PROTOCOL EXECUTION & CLOSED TIMELIKE CURVE (CTC) ANALYSIS

The complete physical trial operates as a 6-step optoelectronic protocol:

1. **Preparation:** Parameter x, laser drive, and time code Y set counting windows.
2. **Energy Switching:** Electrons drop $E_2 \rightarrow E_1$ emitting photons; detector absorption lifts electrons across bandgaps.
3. **Threshold Integration:** Driver A samples noisy signal; Driver B charges C until NPN conducts.
4. **Gate Evaluation:** AND gating enforces dual agreement (preserving one-sided error).
5. **Measurement Latch:** 10µF node filters glitches; 74HC74 latches output Q = SELECT on 10 Hz clock edge.
6. **Repetition:** k independent trials reduce false reject probability down to $(1/2)^k$.

### 7.1 Closed Timelike Curves (CTCs) vs. Electrical Feedback Loops

In the standard hardware implementation, signal flow is strictly **feed-forward** (no feedback path from Q to the drivers or pump). If a feedback wire were introduced from Q back to the input, it would form an ordinary time-delayed electrical loop.

In theoretical physics, Deutsch's model for quantum operations near Closed Timelike Curves (CTCs) enforces a consistency condition on the state $\rho_{CTC}$:

$$\rho_{CTC} = Tr_{other} [ U (\rho_{other} \otimes \rho_{CTC}) U^{\dagger} ]$$

While a CTC resolves grandfather-type paradoxes (e.g. x = NOT x) instantaneously via a 50/50 mixed state density matrix, a physical circuit with an inverted feedback loop (wiring Q-bar to D) resolves state contradiction through a **clock delay of T = 100 ms**, producing a stable 5 Hz output oscillation rather than a temporal fixed-point state.

## 8. SCIENTIFIC INTERPRETATION & CONCLUSION

RP-GHOST unifies physical-to-computational feedback, optoelectronic photon regulation, and formal randomized algorithm verification. The architecture establishes that while physical sensing and optical statistics can yield ultra-low noise deterministic gating and zero-false-accept logic, fundamental computational complexity classes (such as SHA-256d exponential proof-of-work search) remain strictly governed by mathematical difficulty boundaries.